

Project Initialization and Planning Phase

Date	5 June 2026
Team ID	xxxxxx xxxxxx
Project Name	3 Marks
Maximum Marks	

Define Problem Statements :

Floods are among the most devastating natural disasters, claiming thousands of lives and displacing millions every year. Despite their recurring nature, the lack of timely and accurate early-warning systems continues to amplify their destructive impact. Conventional forecasting methods often fall short in predicting floods at the right time, leaving authorities and communities with insufficient opportunity to respond.

This project addresses that gap by building a machine learning-powered flood prediction system trained on historical weather data. Using classification algorithms Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost the system analyses meteorological features such as annual rainfall, cloud visibility, and seasonal rainfall patterns to predict the likelihood of a flood event.

The best-performing model is saved and integrated into a Flask web application, enabling concerned authorities and disaster management teams to monitor flood risk predictions through an intuitive and accessible user interface.